SitePulse

Workforce Attendance & Manpower Control

GPS-verified attendance for every worker, on every project — captured at the gate and in head office at the same moment.

KTC International Contracting

Prepared for Human Resources

Where attendance breaks down

The four failures SitePulse was built to remove

1

Paper and Excel

A day's attendance is collected on paper, typed up later, and consolidated by hand. Mistakes only surface at payroll.

2

No proof of presence

A name written on a sheet does not prove the man was on the project. There is nothing to check the claim against.

3

Transfers go stale

A worker moves from Project A to Project B and keeps counting against the site he left, until head office reassigns him by hand.

4

Month-end scramble

Trade-wise manpower, absentee lists and hours are rebuilt from scratch every month, under time pressure.

Two apps, one live database

The same data, whether you are standing at the gate or sitting in head office

A

Android app — for site

Installed on the timekeeper's or foreman's phone. GPS, badge scanning, works on a normal site phone.

W

Web app — for office

Opens in any browser, no install. Dashboards, reports, roster and user management.

Both run on Google Firebase — the same infrastructure Google runs its own products on. Data is replicated and backed up by Google; a lost or broken site phone loses nothing.

5
user roles, each with its own access

2
apps, one shared database

0
paper attendance sheets required

How a worker is marked present

Three steps, roughly five seconds per man

1

Identify

Scan the QR or barcode on the worker's ID badge, or search him by name or employee number.

2

Verify location

The phone's GPS is checked against the project's geofence before anything is saved.

3

Record

IN or OUT is stored with the time, shift, project, and the name of the person who marked it.

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If the phone is outside the project's geofence, the check-in is refused

The refused attempt is still recorded — who tried, where they were, and how far off site — so a pattern of attempts is visible rather than lost.

ID badges and scanning

The worker's employee number, printed as a code the phone can read

1

Generate

Every worker's employee number becomes a QR code. Nothing to type in, nothing to look up.

2

Print

Twelve badges to an A4 page, each with the worker's name and ID underneath, as one printable file.

3

Scan

Point the phone at the badge. The worker is identified in under a second, with no chance of a mistyped ID.

1

Existing badges still work

The scanner reads ordinary barcodes as well as QR, so badges already issued do not have to be replaced.

2

A badge is never required

Lost his badge, or a new man with none yet? Search by name or employee number and mark him as normal.

Proof the man was there

Location is checked by the system, not asserted by a person

1. **Fake GPS apps are refused outright**
   Android reports a spoofed fix; the check-in is rejected and the attempt recorded.
2. **Every project carries its own geofence**
   Coordinates and radius per project, so a tight compound and a spread-out site differ.
3. **Distance is recorded, not just pass or fail**
   How far from the site centre the phone was, so a borderline check-in can be reviewed.
4. **Off-roster arrivals are flagged**
   Marked at a project other than his roster? Flagged as a site deviation for review.
5. **Office staff have a wider radius**
   Set per site, so office staff are not held to a gate-level fence while the crew is.

A spoofed location is treated as its own kind of refusal, logged the same way an outside-the-fence attempt is.

Checked against the biometric

The one question a geofence cannot settle, answered by the ERP

1

Upload

The ERP 8127 report goes straight into the app — employee ID, name, punch in and out.

2

Compare

Every marked record is matched against the punches for the same day, worker by worker.

3

Record

Who ran the check, when, and what it found is kept — so daily verification is evidenced.

!

Marked present, never punched

Listed for review with the name of whoever marked him. The reverse is caught too: punched but never marked, which costs the worker a day.

i

Honest about its limits

It finds a bad mark afterwards rather than stopping one, and only covers workers a reader reaches — the app says so on screen when coverage is low.

Five roles, scoped by project

Access is decided by role plus the projects that person is assigned to

SA Super Admin

Full access. Creates users, sets roles, configures projects and geofences.

A Admin

Manages the workforce, attendance and reports across their assigned projects.

TK Timekeeper

Owns their site's roster and attendance. Approves incoming worker transfers.

SV Supervisor

Marks attendance, reads manpower and raises transfers for their projects.

ST Staff

Office staff. Marks only their own attendance and applies for their own leave.

Foreman

Merged into Supervisor — same job at KTC. Logins created as Foreman still work, and now read as Supervisor.

A person cannot see, mark or export data for a project they are not assigned to.

Worker moves from Project A to B

Fixed at the gate by the people who can see him, not by head office

BEFORE

The man could be marked present at Project B, but his roster still said Project A. He kept counting against the site he had left until head office reassigned him by hand — often weeks later.

NOW

The foreman who is looking at him raises a transfer. Project B's own timekeeper approves it in one tap and the roster moves across. Head office is not in the loop.

1. **Foreman**
   raises the transfer
2. **Timekeeper**
   approves it
3. **Roster**
   moves to Project B

A foreman can only pull a worker towards a project he is assigned to, and cannot approve his own request.

Month-end in one tap

Every report below is generated as a single Excel workbook, ready to send

1. **Attendance**
   One sheet per project — every IN and OUT with time and shift

2. **Project Summary**
   Headcount and attendance by project, side by side

3. **Trade-wise Summary**
   Manpower by trade, per project or across all projects

4. **Monthly Hours**
   Hours worked per worker across the month

5. **Daily Manpower**
   Present, absent, on leave and late, by project and trade

6. **Absent Report**
   Who was absent, with approved leave and public holidays excluded

Reports respect the same access rules — a timekeeper's export contains their projects only, never the whole company.

The whole workforce on one screen

Live figures for any date and any project, without waiting for a report

- Present
- Absent
- On Leave
- Late
- Checked Out
- Blocked
- Total Staff
- KTC
- Supplier
- Active Projects
- Total Projects

Eleven live figures, illustrated above. Values update as attendance is marked.

?

Worker Locator

Type a name or ID and see which project he was on for any chosen date.

!

Site Deviations

Anyone marked at a project other than their roster, listed for review the same day.

Access is enforced by the database

Hiding a button is a convenience. It is never the security boundary.

1. **Rules live on Google's servers**
   Even a modified app cannot read another project's attendance — the request is refused before it reaches the data.

2. **Nobody can promote themselves**
   A user cannot write their own role or site list. Only Super Admin can, and every change is stamped with who made it.

3. **Attendance is never deleted**
   Corrections are recorded as edits, so the original entry and the correction both survive for audit.

56

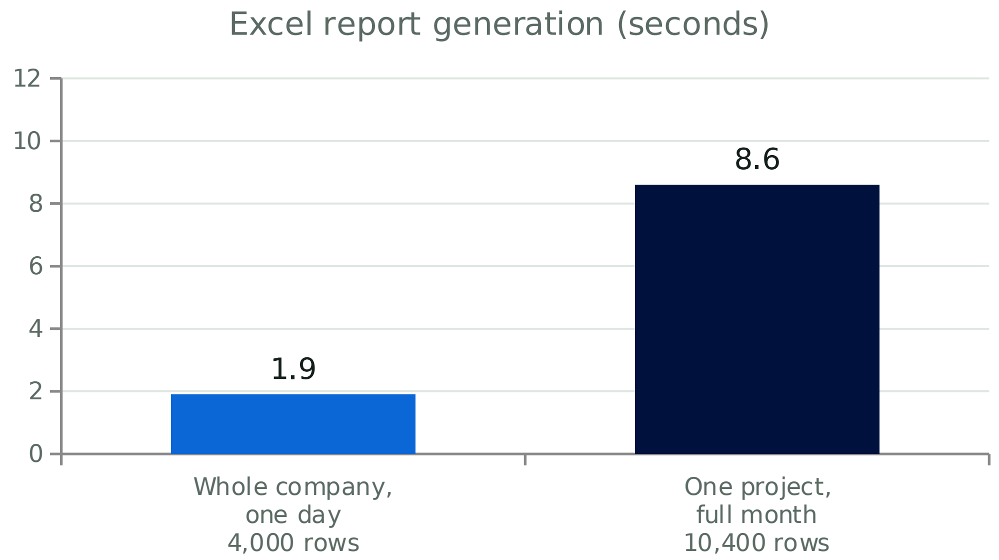
automated security tests

Each one checks a permission that must be granted — and one that must be refused. All 56 pass.

Tests are re-run against a Firebase emulator before every release.

Measured for 4,000 workers

Report generation timed on the standard build, not estimated



1. **Attendance is written per project, per day**
   The system never has to load the whole company at once, so adding projects does not slow the existing ones down.

2. **Every phone loads only its own projects**
   A foreman's phone fetches his site's crew, not 4,000 records.

3. **Growth is priced per record, not per user**
   Adding a site or a supervisor costs nothing extra by itself.

Measured on the release build with an Android-sized memory limit. Both reports completed well inside it.

What this changes for HR

Six outcomes, in the order HR feels them

1 Payroll you can defend

Every paid day traces back to a GPS-verified record naming the person who marked it.

2 Month-end in minutes

Trade-wise, absentee and hours reports generate on demand instead of being rebuilt by hand.

3 Accurate cost allocation

Transfers move the roster the same day, so a worker is charged to the project he actually worked on.

4 Leave handled in the system

Staff apply, admin approves, and approved leave is automatically excluded from absence.

5 Disputes settled by record

Worker Locator answers 'where was he on the 14th?' in seconds, for any past date.

6 Nothing lost with a phone

Data lives in Google's cloud, not on the device. A broken or stolen phone loses no attendance.

Running cost and what comes next

No licence fees, no per-user charge, no server to buy

COST

There is no software licence and no per-user fee. The only cost is Google's charge for data stored and read, which scales with how much the system is actually used.

At 4,000 workers this is projected to sit well inside a USD 100 per month budget.

Already live today

GPS check-in, QR badges, ERP biometric cross-check, five roles, site scoping, transfers, leave, holidays, dashboards, Excel reports and full backup.

Cost figure is a projection based on expected usage, not a quoted price.

ALREADY PLANNED

1. **Broadcast notifications**
   Alerts for transfers, arrivals and leave. The daily biometric reminder already works.
2. **Photo on check-in**
   An optional photo stored with the attendance record.
3. **Power BI / Excel reporting**
   Live connection for HR's own analysis, without touching the app.
4. **ERP roster sync**
   Roster arriving automatically from the ERP instead of an Excel upload.

Built, tested, ready to roll out

Both apps are complete and running against the live database. What remains is a pilot on one project, then a staged rollout across the rest.

4,000
workers supported

5
roles

56
security tests passing